

Clustering

Rishi Shanthan Bhagavatham

2024-09-10

12.5.3 *K-Means Clustering*

```
set.seed (2)
x <- matrix(rnorm(50 * 2), ncol = 2)
x[1:25, 1] <- x[1:25, 1] + 3
x[1:25, 2] <- x[1:25, 2] - 4
```

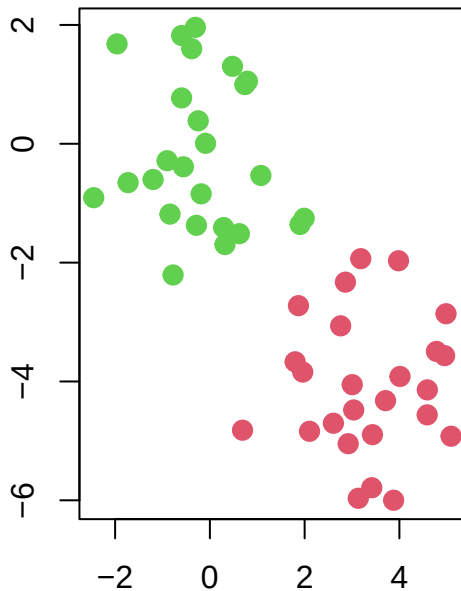
```
km.out <- kmeans(x, 2, nstart = 20)
```

```
km.out$cluster
```

```
## [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2
## [39] 2 2 2 2 2 2 2 2 2 2 2 2 2
```

```
par(mfrow = c(1, 2))
plot(x, col = (km.out$cluster + 1),
main = "K-Means Clustering Results with K = 2", xlab = "", ylab = "", pch = 20, cex = 2)
```

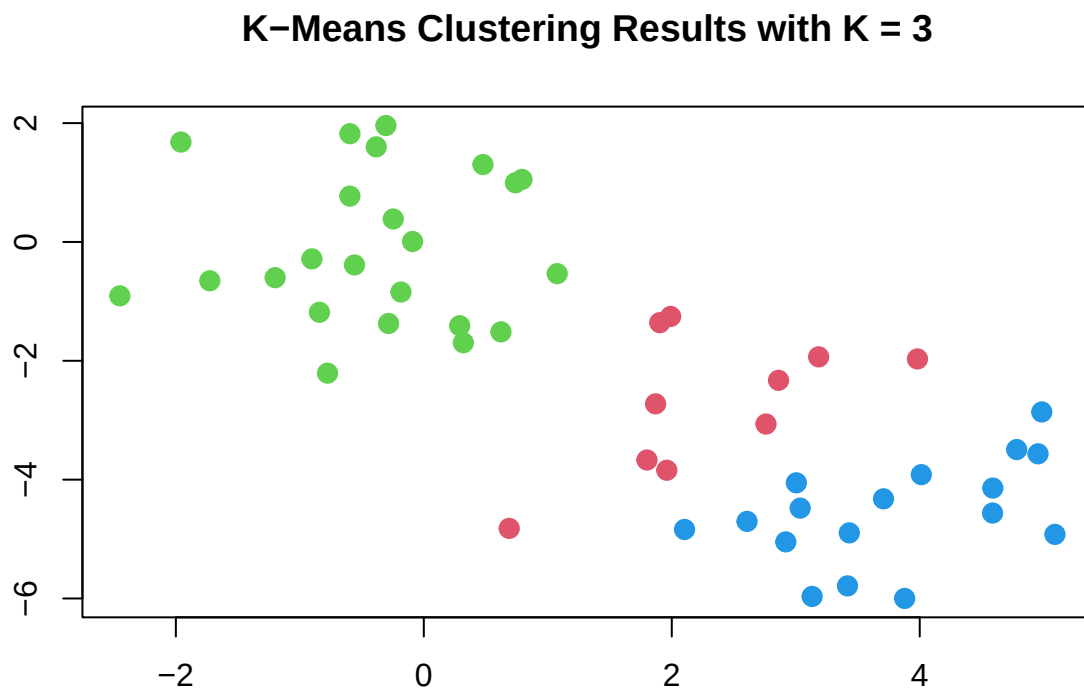
K-Means Clustering Results with K



```
set.seed (4)
km.out<-kmeans(x, 3, nstart = 20)
km.out

## K-means clustering with 3 clusters of sizes 10, 23, 17
##
## Cluster means:
##      [,1]      [,2]
## 1  2.3001545 -2.69622023
## 2 -0.3820397 -0.08740753
## 3  3.7789567 -4.56200798
##
## Clustering vector:
## [1] 3 1 3 1 3 3 3 1 3 1 3 1 3 1 3 1 3 3 3 3 1 3 3 3 2 2 2 2 2 2 2 2 2 2
## [39] 2 2 2 2 2 1 2 1 2 2 2 2
##
## Within cluster sum of squares by cluster:
## [1] 19.56137 52.67700 25.74089
## (between_SS / total_SS =  79.3 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```
plot(x,col = (km.out$cluster + 1),main = "K-Means Clustering Results with K = 3", xlab = "", ylab = "", pch = 1)
```



```
set.seed(4)
km.out<-kmeans(x, 3,nstart = 1)
km.out$tot.withinss
```

```
## [1] 104.3319
```

```
km.out<-kmeans(x, 3,nstart = 20)
km.out$tot.withinss
```

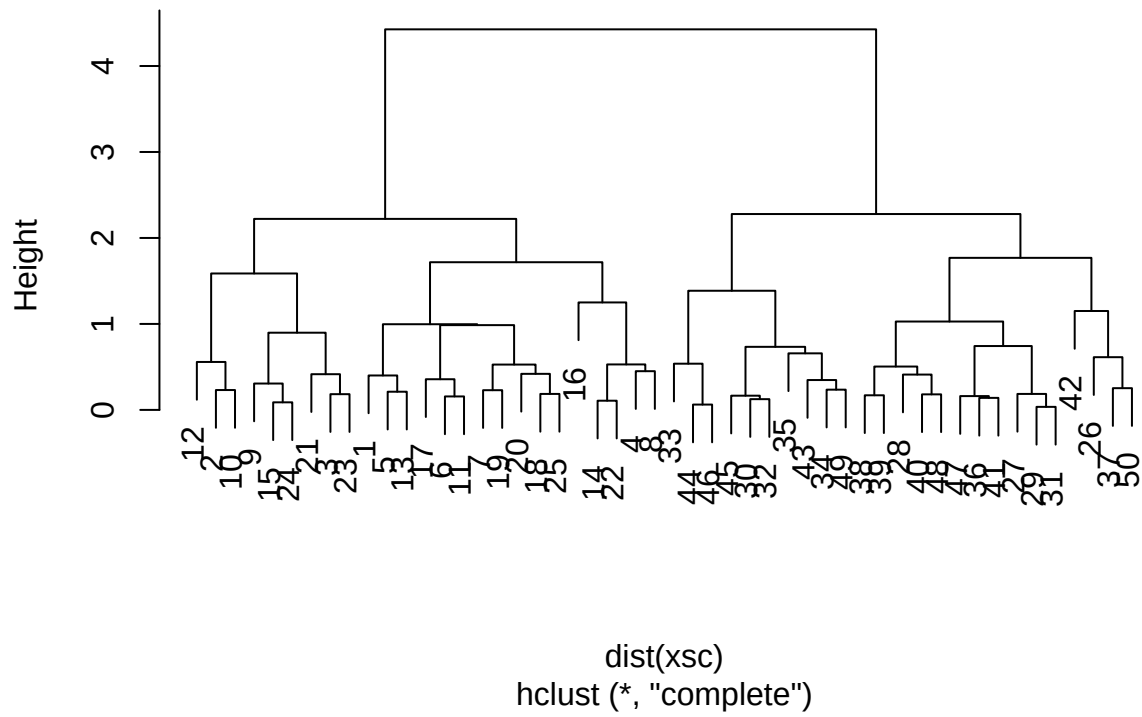
```
## [1] 97.97927
```

Hierarchical Clustering

```
hc.complete <- hclust(dist(x), method = "complete")
hc.average <- hclust(dist(x), method = "average")
hc.single <- hclust(dist(x), method = "single")
```

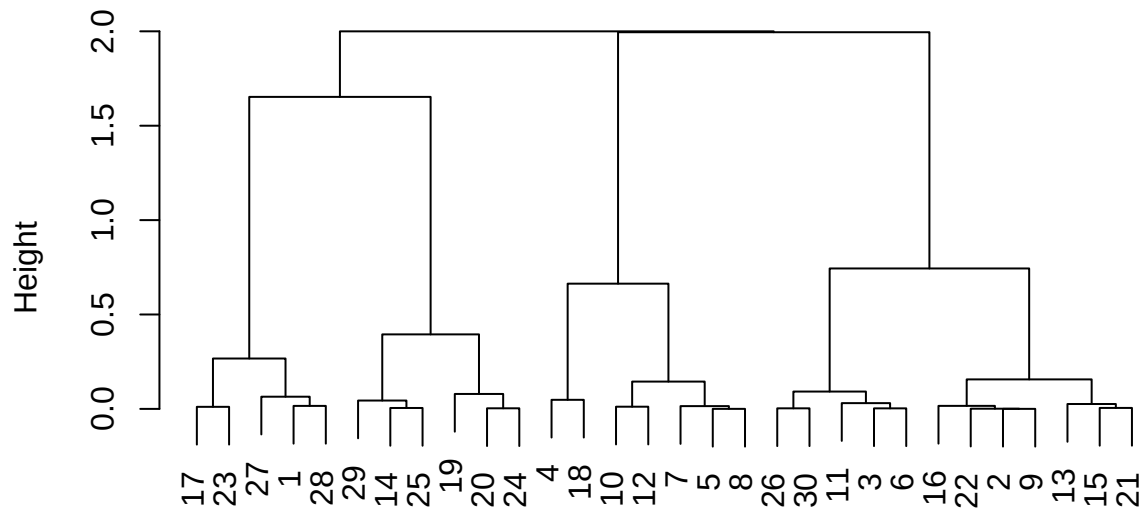
```
par(mfrow = c(1,3))
plot(hc.complete,main = "Complete Linkage",xlab = "", sub = "",cex=.9)
plot(hc.average,main = "Average Linkage", xlab = "", sub="",cex =.9)
plot(hc.single,main = "Single Linkage", xlab = "", sub = "", cex = .9)
```


Hierarchical Clustering with Scaled Features



```
x <-matrix(rnorm(30 * 3), ncol = 3)
dd <-as.dist(1-cor(t(x)))
plot(hclust(dd, method = "complete"),
main = "Complete linkage with Correlation-Based Distance", xlab="", sub="")
```

Complete linkage with Correlation-Based Distance



```
# This step is to load the USArrests data
data("USArrests")
head(USArrests)
```

```
##           Murder Assault UrbanPop Rape
## Alabama      13.2    236         58 21.2
## Alaska       10.0    263         48 44.5
## Arizona       8.1    294         80 31.0
## Arkansas      8.8    190         50 19.5
## California    9.0    276         91 40.6
## Colorado      7.9    204         78 38.7
```

```
#The Mean of each Feature
apply(USArrests , 2 , mean)
```

```
##      Murder  Assault UrbanPop      Rape
##      7.788  170.760  65.540   21.232
```

```
#The Variance of each Feature
apply(USArrests , 2 , var)
```

```
##      Murder      Assault      UrbanPop      Rape
##  18.97047 6945.16571  209.51878  87.72916
```

lab #9 Consider the USArrests data. We will now perform hierarchical clustering on the states.

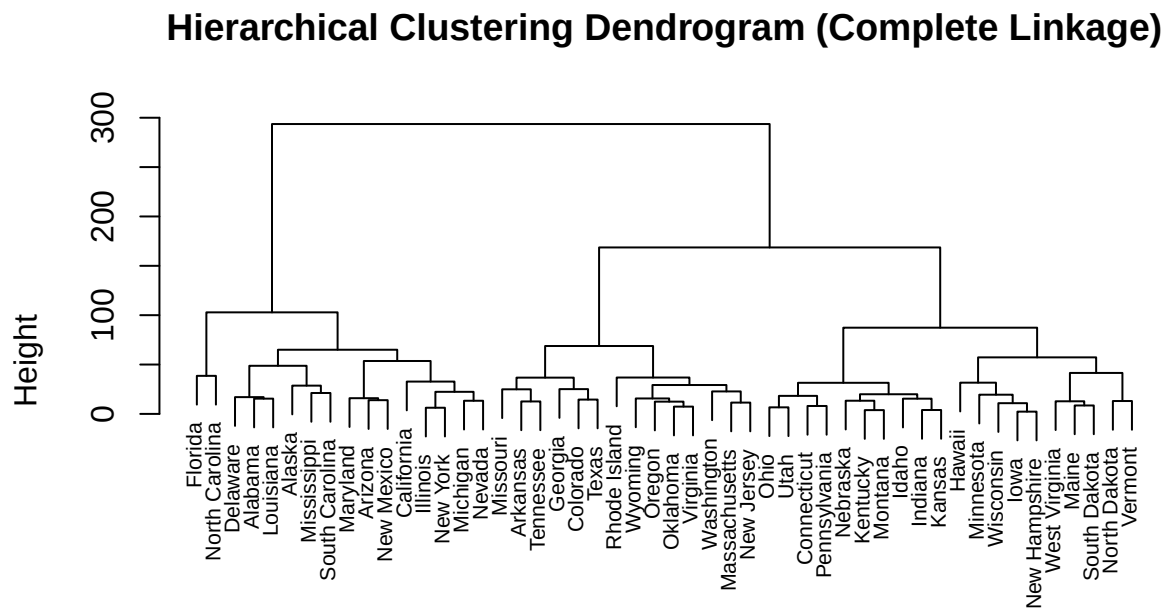
- (a) Using hierarchical clustering with complete linkage and Euclidean distance, cluster the states.

```
#First we need to compute the Distance Matrix
dist_matrix <- dist(USArrests, method = "euclidean")
```

```
#Perform hierarchical clustering using complete linkage
hc_complete <- hclust(dist_matrix, method = "complete")
```

```
#Then Plot the dendrogram
```

```
plot(hc_complete, main = "Hierarchical Clustering Dendrogram (Complete Linkage)", xlab = "", sub = "",
```



- (b) Cut the dendrogram at a height that results in three distinct clusters. Which states belong to which clusters?

```
#First cut the dendrogram to form three clusters
```

```
clusters <- cutree(hc_complete, k = 3)
```

```
#Create a data frame with the cluster assignments
```

```
clustered_states <- data.frame(State = rownames(USArrests), Cluster = clusters)
```

```
clustered_states <- clustered_states[order(clustered_states$Cluster), ]
```

```
#Display the states in each cluster
cluster_list <- split(clustered_states$State, clustered_states$Cluster)
cluster_list
```

```
## $'1'
## [1] "Alabama"      "Alaska"      "Arizona"     "California"
## [5] "Delaware"     "Florida"     "Illinois"    "Louisiana"
## [9] "Maryland"     "Michigan"    "Mississippi" "Nevada"
## [13] "New Mexico"   "New York"    "North Carolina" "South Carolina"
##
## $'2'
## [1] "Arkansas"     "Colorado"    "Georgia"     "Massachusetts"
## [5] "Missouri"     "New Jersey"  "Oklahoma"    "Oregon"
## [9] "Rhode Island" "Tennessee"   "Texas"       "Virginia"
## [13] "Washington"   "Wyoming"
##
## $'3'
## [1] "Connecticut"  "Hawaii"     "Idaho"       "Indiana"
## [5] "Iowa"         "Kansas"     "Kentucky"    "Maine"
## [9] "Minnesota"    "Montana"    "Nebraska"    "New Hampshire"
## [13] "North Dakota" "Ohio"       "Pennsylvania" "South Dakota"
## [17] "Utah"         "Vermont"    "West Virginia" "Wisconsin"
```

- (c) Hierarchically cluster the states using complete linkage and Euclidean distance, after scaling the variables to have standard deviation one.

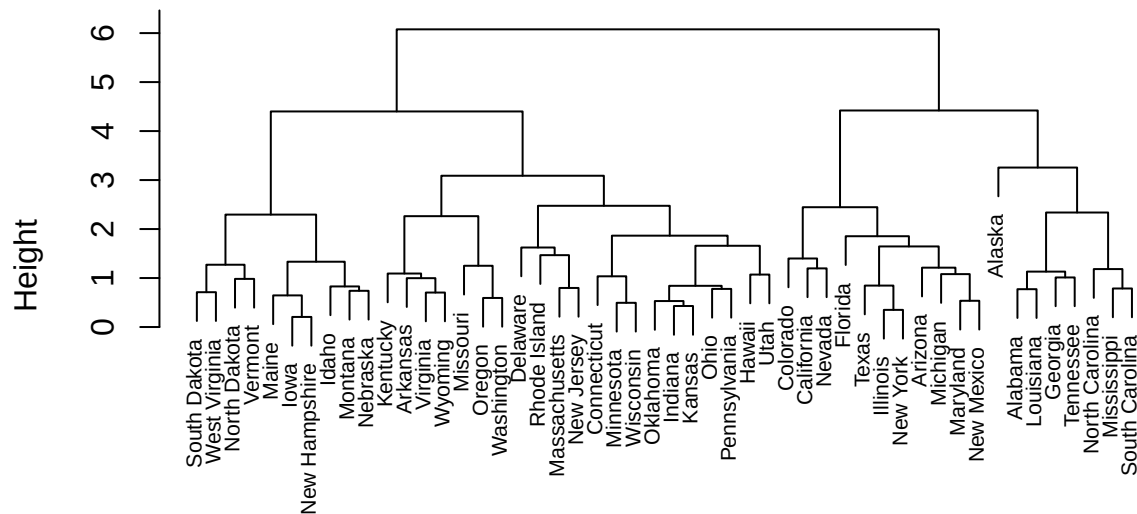
```
#Scale the variables to have standard deviation one
scaled_data <- scale(USArrests)

#Then we need to compute the Distance Matrix
dist_matrix_scaled <- dist(scaled_data, method = "euclidean")

#Perform hierarchical clustering using complete linkage
hc_complete_scaled <- hclust(dist_matrix_scaled, method = "complete")

#Then Plot the dendrogram
plot(hc_complete_scaled, main = "Hierarchical Clustering Dendrogram (Complete Linkage, Scaled Data)", xlab = "State", ylab = "Distance",
```


Hierarchical Clustering Dendrogram (Complete Linkage, Scaled Data)



- (d) What effect does scaling the variables have on the hierarchical clustering obtained? In your opinion, should the variables be scaled before the inter-observation dissimilarities are computed? Provide a justification for your answer.

```
#Lets compare the dendrograms of a three cluster group from the scaled data for a reference
#So that cut the dendrogram of the scaled data to form three clusters
clusters_scaled <- cutree(hc_complete_scaled, k = 3)

#Create a data frame with the cluster assignments
clustered_states_scaled <- data.frame(State = rownames(USArrests), Scaled_Cluster = clusters_scaled)
clustered_states_scaled <- clustered_states_scaled[order(clustered_states_scaled$Scaled_Cluster), ]

# Display the states in each cluster for scaled data
scaled_cluster_list <- split(clustered_states_scaled$State, clustered_states_scaled$Scaled_Cluster)
scaled_cluster_list

## $'1'
## [1] "Alabama"      "Alaska"      "Georgia"     "Louisiana"
## [5] "Mississippi"  "North Carolina" "South Carolina" "Tennessee"
##
## $'2'
## [1] "Arizona"      "California" "Colorado"    "Florida"    "Illinois"
## [6] "Maryland"     "Michigan"   "Nevada"      "New Mexico" "New York"
## [11] "Texas"
##
```

```
## $'3'
## [1] "Arkansas"      "Connecticut"   "Delaware"      "Hawaii"
## [5] "Idaho"         "Indiana"       "Iowa"          "Kansas"
## [9] "Kentucky"      "Maine"         "Massachusetts" "Minnesota"
## [13] "Missouri"      "Montana"       "Nebraska"      "New Hampshire"
## [17] "New Jersey"    "North Dakota"  "Ohio"          "Oklahoma"
## [21] "Oregon"        "Pennsylvania"  "Rhode Island"  "South Dakota"
## [25] "Utah"          "Vermont"       "Virginia"      "Washington"
## [29] "West Virginia" "Wisconsin"     "Wyoming"
```

#Compare the original and scaled clustering results

```
comparison <- merge(clustered_states, clustered_states_scaled, by = "State")
comparison <- comparison[order(comparison$Cluster), ]
comparison
```

##	State	Cluster	Scaled_Cluster
## 1	Alabama	1	1
## 2	Alaska	1	1
## 3	Arizona	1	2
## 5	California	1	2
## 8	Delaware	1	3
## 9	Florida	1	2
## 13	Illinois	1	2
## 18	Louisiana	1	1
## 20	Maryland	1	2
## 22	Michigan	1	2
## 24	Mississippi	1	1
## 28	Nevada	1	2
## 31	New Mexico	1	2
## 32	New York	1	2
## 33	North Carolina	1	1
## 40	South Carolina	1	1
## 4	Arkansas	2	3
## 6	Colorado	2	2
## 10	Georgia	2	1
## 21	Massachusetts	2	3
## 25	Missouri	2	3
## 30	New Jersey	2	3
## 36	Oklahoma	2	3
## 37	Oregon	2	3
## 39	Rhode Island	2	3
## 42	Tennessee	2	1
## 43	Texas	2	2
## 46	Virginia	2	3
## 47	Washington	2	3
## 50	Wyoming	2	3
## 7	Connecticut	3	3
## 11	Hawaii	3	3
## 12	Idaho	3	3
## 14	Indiana	3	3
## 15	Iowa	3	3
## 16	Kansas	3	3
## 17	Kentucky	3	3
## 19	Maine	3	3

## 23	Minnesota	3	3
## 26	Montana	3	3
## 27	Nebraska	3	3
## 29	New Hampshire	3	3
## 34	North Dakota	3	3
## 35	Ohio	3	3
## 38	Pennsylvania	3	3
## 41	South Dakota	3	3
## 44	Utah	3	3
## 45	Vermont	3	3
## 48	West Virginia	3	3
## 49	Wisconsin	3	3

It's clear that the clusters are different among the scaled and the regular data. Generally, scaling the variables ensures that each variable contributes equally to the distance calculations. Without scaling, variables with larger variances can dominate the clustering process, potentially leading to misleading clusters. In the USArrests dataset, the variables have different ranges, so scaling is generally recommended to prevent any single variable from having an outsized influence on the clustering outcome.